Office of Institutional Research & Effectiveness
NSSE Survey Analysis

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Executive Summary

This report analyzes the National Survey of Student Engagement (NSSE) survey data collected at Trinity University in 2022 and 2025 to examine how student engagement evolves across the undergraduate experience and what themes influence that development. Three research questions were investigated: (1) whether student perceptions of Academic Challenge (AC), Learning with Peers (LP), Experiences with Faculty (EF), and Campus Environment (CE) change from freshman to senior year; (2) whether changes in AC and EF differ between students who participated in high impact practices (internships and/or study abroad) and those who did not; and (3) whether perceptions of CE and EF differ among Trinity's four academic schools. To address RQ1, a repeated-measures MANOVA and follow-up ANOVAs were conducted on a matched sample of students who completed the survey in both their freshman and senior year. RQ2 was examined using linear mixed effects models to account for the repeated-measures structure of the data while testing whether high-impact practices participation moderated engagement trajectories over time. RQ3 was addressed using one-way ANOVAs on a pooled cross-sectional dataset combining both survey cohorts.

Results indicate that perceptions relating to AC and EF increased significantly from freshman to senior year, with EF showing the largest growth, while CE declined significantly. Furthermore, students who participated in high-impact practices during their first year at Trinity had higher AC scores as freshmen, though non-participants largely caught up by senior year, suggesting a convergence effect. Finally, CE and EF perceptions across Trinity's four academic schools were consistent, indicating that institutional-level approaches to improving these outcomes are appropriate. Overall, the findings suggest EF and early experiential learning participation to be the most actionable decisions for institutional improvement.

Organizational Information

Trinity University's Office of Institutional Research and Effectiveness (TUIR) is responsible for institutional data management, reporting, and assessment. The term effectiveness is of key importance to the office. The idea of effectiveness arose 80 years ago with a focus on rudimentary basics, such as only reporting enrollment and faculty information. Since then, the term has expanded to support institutional accreditation, strategic planning, and decision-making.

TUIR provides data and analysis to key stakeholders, including members of the Trinity Leadership team, such as Deans, Provost, Vice Presidents, and the Board of Trustees. The office also adheres to federal and state reporting requirements and supports external reporting to organizations that rely on institutional data, such as the Princeton Review and US News. Reports are also made available to prospective students and their families to help guide college decision-making.

Since institutional effectiveness is a core responsibility of TUIR, the office mainly engages in assessing whether the University meets stated outcomes and accreditation standards (SACSCOC) and demonstrates continuous improvement. To support this work, TUIR administers and analyzes institutional surveys, including the NSSE.

At Trinity, the NSSE is administered every 3 years to first-year students and graduating seniors. NSSE measures student engagement through 10 composite engagement indicators and participation in high-impact practices, such as experiential learning, internships, and study abroad. Since engagement is strongly correlated with student retention, academic success, and graduation rates, NSSE's data provides an important source of evidence for evaluating institutional performance. Results directly support leadership discussions and shape decision-making in academic and student affairs.

Operational Background: NSSE Core Survey

The NSSE core survey measures student engagement using Likert-scale responses, where students respond with the extent to which they engage in behaviors during the academic year. These individual survey items are then aggregated into composite measures known as Engagement Indicators, which capture broader dimensions of student engagement. In total, there are ten Engagement Indicators, which are organized into four overarching categories:

Table 1. Overview of engagement themes and component indicators

Engagement Theme	Component Indicators	Description
Academic Challenge (AC)	Higher-Order Learning, Reflective & Integrative Learning, Learning Strategies, Quantitative Reasoning	Measures the extent to which students engage in rigorous, higher-level cognitive work through analysis, integration of ideas, effective learning strategies, and the application of quantitative reasoning to real-world problems.
Learning with Peers (LP)	Collaborative Learning, Discussions with Diverse Others	Measures the extent to which students engage in collaborative learning and meaningful interactions with peers, including working together on academic tasks and engaging in discussions with individuals from diverse backgrounds.
Experiences with Faculty (EF)	Student–Faculty Interaction, Effective Teaching Practice	Measures the extent to which students engage with faculty through academic and career-related interactions and benefit from effective teaching practices that support learning and development.
Campus Environment (CE)	Quality of Interactions, Supportive Environment	Measures the extent to which students perceive their institution as supportive and inclusive, reflected in the quality of their interactions with campus stakeholders and the availability of resources that support their academic, social, and personal well-being.

Each of the component Engagement Indicators, as well as their relevance, is discussed in detail in the data section.

Beyond engagement indicators, the survey also touches on High-Impact Practices (HIP), which include extracurricular activities related to learning outside the classroom and professional development, such as internships, study abroad, research, or service learning. From these survey responses, NSSE derives summary measures that allow comparisons between student groups. The survey also includes a number of summary and demographic measures, such as estimated number of study hours, first-generation status, class year, and major, to allow for segmentation of respondents.

It is important to note that NSSE also provides survey weights in order for data to be adjusted for response patterns relating to demographic variables such as sex and enrollment status. These weights allow institutional reports to better reflect their undergraduate population. Since this investigation will examine differences between first and fourth-year students, the structure of the NSSE survey, including identifiers, composites, and derived measures, is central to the analytical strategy. The standardized nature of the response data ensures that comparability across cohorts is maintained.

Research Questions

The NSSE dataset provides a valuable opportunity to examine how student engagement evolves over the course of a student's undergraduate experience. Institutional reports usually summarize student engagement at one point in time; however, the availability of matched freshman and senior survey responses allows for an examination of how student engagement changes throughout the undergraduate experience. Understanding these patterns is important for institutional assessment because engagement indicators are strongly associated with student success outcomes, such as academic performance, retention, and graduation rates.

This study, therefore, aims to evaluate how key engagement themes measured by NSSE develop from freshman to senior year and to explore whether these patterns vary across different student experiences and academic contexts. The first research question (RQ1) serves as the primary and most general investigation, providing an overall assessment of how engagement changes over time. The subsequent questions (RQ2 and RQ3) provide more focused analyses that examine specific factors associated with engagement patterns.

RQ1 and RQ2 employ a repeated-measures design, using matched freshman and senior responses to examine changes in engagement over time for the same students. In contrast, RQ3 uses a pooled cross-sectional approach, combining observations from the 2022 and 2025 surveys to examine differences in engagement perceptions across academic schools. Together, these research questions provide a structured approach for understanding both general trends in engagement and the specific factors that may influence them. The research questions examined in this study are as follows:

- RQ1. Do student perceptions of Academic Challenge, Learning with Peers, Experiences with Faculty, and Campus Environment change from freshman year to senior year?
- RQ2. Do changes in Academic Challenge and Experiences with Faculty from freshman to senior year differ between students who participated in study abroad and/or internships and those who did not?
- RQ3. Do perceptions of Campus Environment and Experiences with Faculty differ among the four academic schools of Arts & Humanities (A&H), Social Sciences and Civic Engagement (SS&CE), Semmes School of Science (SSS), and Neidorff School of Business (NSB)?

Data Overview

The dataset used for this analysis consists of two separate tables derived from responses to the NSSE Core Survey administered at Trinity University. The first dataset contains responses from freshman and senior students in 2022, while the second dataset contains responses with freshman and senior students in 2025. Each table was provided in a spreadsheet format and was converted to a Comma-Separated Values file to be utilized in our analysis. Each dataset included a unique identifier labeled FakeID, which allows responses from the same student to be matched across the two survey administrations in 2022 and 2025.

In total, the first dataset contains 298 columns and 553 rows, while the second dataset contains 230 columns and 562 rows. The differences in columns between the two administrations is attributed to a set of supplementals questions only present in the 2022 administration and a set of gender identity/sexual orientation questions only present in 2025 data. The columns are divided into 2 main sections: survey responses and demographic information. The survey responses section contains a column for each question included in the NSSE core survey, along with a numerical value representing the Likert score provided by the student. There is also a segment of composite columns that aggregate the Likert scores given by students for each question that corresponds to each of the 10 engagement indicators:

Table 2. Engagement Indicator Descriptions

Engagement Indicator	Description
Higher-Order Learning (HO)	Measures the extent to which students engage in higher-level cognitive tasks such as analysis, synthesis, and evaluation.
Reflective and Integrative Learning (RI)	Captures how often students connect ideas from different sources and reflect on their learning experiences.
Learning Strategies (LS)	Measures the degree to which students use effective study strategies and organizational techniques.
Quantitative Reasoning (QR)	Evaluates how frequently students apply numerical reasoning and data analysis in their coursework.
Collaborative Learning (CL)	Measures the frequency of working with peers on academic tasks.
Discussions with Diverse Others (DD)	Captures engagement with individuals from different backgrounds or perspectives.
Student–Faculty Interaction (SF)	Measures interactions between students and faculty members related to academic work.
Effective Teaching Practices (ET)	Evaluates students' perceptions of teaching quality and instructional effectiveness.
Quality of Interactions (QI)	Measures the quality of students' interactions with faculty, advisors, and other institutional members.
Supportive Environment (SE)	Captures perceptions of how supportive the institution is toward student success and well-being.

Each engagement indicator is reported on an individual basis by students, with an institutional score being given based on mean student responses. The demographic information segment of columns includes a variety of columns ranging from whether the students have completed internship and study abroad experiences to their race, gender, age, and disability status.

Data Cleaning and Reshaping

Given the structure of the research questions, data cleaning was done based on the variables relevant to each analysis rather than applying a single global cleaning procedure. For example, RQ1 examined changes in the four engagement themes (AC, EF, LP, CE) and therefore required cleaning of the 10 composite engagement indicator variables that comprise the engagement themes. RQ2 and RQ3 introduced additional categorical variables, such as participation in high-impact practices (constructed from intern and abroad variables) and academic school classification, and therefore required additional variable-specific cleaning procedures. This research question-driven approach allowed the dataset to retain the maximum number of observations while ensuring that each analysis used appropriately prepared variables. Missing observations that were not removed during global cleaning were handled as needed within the specific analyses (e.g., RQ3 when academic school assignment was required).

RQ1. Do student perceptions of Academic Challenge, Learning with Peers, Experiences with Faculty, and Campus Environment change from freshman year to senior year?

The data preprocessing phase for RQ1 consisted of 3 operations: cleaning, matching, and grouping of variables. The raw data set included several placeholders for missing fields, which included entries such as "NULL" and "#NULL". These entries were standardized and replaced with true *null* values to ensure the data could be processed for ANOVA testing in Python. Each dataset also included a primary key field labeled "FakeID", which represented the individual respondents. To ensure that each FakeID appeared only once in both the freshman and senior dataset, duplicate records were removed, keeping the response which contained the greatest number of non-missing values across the engagement indicators. This ensured that the most complete response was preserved and that the final dataset contained a single row for each student in both their freshman and senior surveys.

The totality of the NSSE response data contained a multitude of variables that were not relevant to RQ1 specifically. As such, it was necessary to produce a sub-data frame that contained only the relevant columns. Accordingly, a new data frame was created with FakeID, class, and the ten engagement indicator composite columns.

The specific columns chosen were the composite columns that represented the sum of response scores across all questions related to that specific indicator. Once the engagement indicators were isolated, they were grouped into the following themes:

Table 3. Grouping of engagement indicators

Engagement Theme	Engagement Indicators
AC	HO, RI, LS, QR
LP	CL, DD
EF	SF, ET
CE	QI, SE

These themes were created by averaging the grouped engagement indicators. After cleaning and filtering each dataset individually, the two data frames were merged using FakeID as the common key, ensuring each record was matched from freshman year to senior year. An inner join was used to ensure only students who completed the survey in both freshman year and senior year would appear in the final data set to preserve the integrity of the repeated measures approach. The resulting data frame was composed of 120 rows.

The cleaned and reshaped dataset was therefore organized in a format suitable for analysis of variance (ANOVA), which was used to evaluate differences in the four themes over time.

RQ2. Do changes in Academic Challenge and Experiences with Faculty from freshman to senior year differ between students who participated in study abroad and/or internships and those who did not?

The preprocessing steps for RQ2 were built upon the cleaned and merged dataset created for RQ1. Since this research question examined how changes in engagement differed based on participation in experiential learning opportunities, additional variables related to internships and study abroad participation were required.

First, the internship (intern) and study abroad (abroad) variables were examined in both freshman and senior datasets. Like the cleaning procedure described previously, placeholder entries such as “NULL” and “#NULL” were standardized and converted to true null values to ensure compatibility with numerical processing in Python. Because the goal of this research question was to distinguish between students who participated in experiential learning opportunities and those who did not, these variables were coded as binary indicators representing participation status.

Using the cleaned internship and study abroad fields, two new categorical variables were constructed: experiential_learning_freshman and experiential_learning_senior. These variables identified whether a student reported participating in either an internship or a study abroad experience during the respective survey administration. Students who reported participation in at least one of these activities were coded as having an experiential learning experience, while those who reported neither were coded as non-participants. This grouping allowed the analysis to compare engagement trends between students who participated in high-impact practices and those who did not.

Following the creation of these variables, the dataset was filtered to retain only the engagement indicators relevant to RQ2. Specifically, the engagement themes AC and EF were included, as these themes were the focus of the research question. As described in RQ1, AC was constructed from the HO, RI, LS, and QR indicators, while EF was constructed from the SF and ET indicators. Composite scores for these themes were calculated using the average of the corresponding indicators.

Because the research question focused on changes over time, the merged dataset containing both freshman and senior responses was retained. Each student, therefore, appeared once in the dataset with separate engagement scores for freshman and senior year, along with their experiential learning classification. This structure allowed for comparison of engagement changes between students who participated in experiential learning opportunities and those who did not. The resulting dataset contained 119 rows and 7 columns.

RQ3. Do perceptions of CE and EF differ among the four academic schools (A&H, SS&CE, SSS, NSB)?

The data preprocessing phase for RQ3 differed from the previous two research questions in that it did not require a repeated measures structure. While RQ1 and RQ2 relied on matching respondents across the 2022 and 2025 surveys using FakeID, RQ3 examined differences in engagement perceptions across academic schools and therefore used the full set of available observations from both survey years. As such, the preprocessing for RQ3 focused primarily on preparing the relevant categorical variables and engagement outcomes rather than matching records across time.

The primary categorical variable used in this analysis was **MAJfirstcol**, which represented the respondent's primary major classification. The variable originally consisted of 11 disciplinary categories defined by the NSSE dataset. Each of these categories was recoded to align with Trinity University's four academic schools: A&H, SS&CE, SSS, and NSB. A small number of cases were initially coded as "11", representing "Other." These cases were reviewed individually and were assigned to the SSS category because they were primarily computer related.

A secondary categorical variable, **MAJsecondcol**, was also examined to determine whether missing values in MAJfirstcol could be supplemented. However, cases with missing primary major classifications did not contain valid secondary major responses. Instead, these entries were coded as "-9," indicating that the respondent did not receive the survey question regarding a second major. As a result, MAJsecondcol did not provide any additional information that could be used to recover missing primary major classifications.

Additionally, a number of students reported secondary majors that belonged to different academic schools than their primary major. More specifically, there were 73 such instances in the 2022 dataset and 71 in the 2025 dataset. Among these cases, the most common pairing involved a primary major in SS&CE and a secondary major in A&H. To ensure that each observation was assigned to a single academic division, analyses were conducted using the primary major classification (MAJfirstcol). This approach aligned with standard institutional reporting practices, where the primary major reflected the student's primary academic field.

The engagement outcomes for RQ3 were derived from the same NSSE engagement indicator composite variables used in the previous analyses. Specifically, the themes CE and EF were constructed from the composite indicators QI and SE, along with SF and ET, respectively.

As with RQ1, these themes were calculated using the average of their respective engagement indicators to ensure consistency in scale.

Missing values for engagement indicators were not removed during the initial preprocessing stage in order to preserve the full analytic sample. This approach was consistent with the data cleaning procedure used for RQ1. Instead, missing observations were handled at the analysis stage as required by ANOVA. Similarly, undeclared (999) and missing major classifications were retained in the full dataset during preprocessing but were excluded from analyses requiring assignment to a specific academic school.

The resulting dataset, therefore, retained the full set of observations from both survey years while ensuring that all variables relevant to RQ3 were appropriately cleaned, recoded, and structured for subsequent analysis. The resulting dataset contained 1,115 rows and 345 columns.

Descriptive Statistics

Descriptive statistics were calculated using the same research question-driven approach described in the data cleaning process. Since each research question relies on different sets of variables, descriptive summaries were generated only for the variables relevant to each analysis. This approach ensured that the descriptive statistics accurately reflected the appropriate analytical sample.

RQ1. Do student perceptions of Academic Challenge, Learning with Peers, Experiences with Faculty, and Campus Environment change from freshman year to senior year?

Table 4 shows the descriptive statistics and correlations among the four academic engagement themes for freshman students. The mean scores indicate that CE has the highest average score with M = 41.44 (SD = 9.27); this shows that first-year students generally view

Trinity as a positive environment. LP also shows a relatively high mean with M = 38.11 (SD = 11.49), indicating that there is frequent collaborative learning and peer interaction. In contrast, EF has the lowest mean with M = 31.90 (SD = 9.72). The correlation matrix shows positive relationships among all the themes; the strongest association is between CE and EF (r = 0.56). This suggests that students who reported more frequent interactions with faculty perceived the campus environment to be more positive. Values on the diagonal represent the square root of the Average Variance Extracted (AVE) for each theme which should exceed all off-diagonal correlations in the same row and column to confirm that each theme measures a distinct construct.

Table 4. Descriptive Statistics and Correlations of Academic Themes for First Years

Construct	Mean	SD	1	2	3	4
1. AC	37.22	9.58	**0.76**			
2. LP	38.11	11.49	0.49	**0.80**		
3. EF	31.90	9.72	0.53	0.38	**0.76**	
4. CE	41.44	9.27	0.41	0.29	0.56	**0.8**

Note: AC = Academic Challenge; LP = Learning with Peers; EF = Experiences with Faculty; CE = Campus Environment.

Table 5 shows the descriptive statistics, reliability measures, and correlations among the four academic engagement themes for Senior students. Compared to first-year students, Seniors reported higher mean scores across almost all four themes. The mean scores indicate that AC has the highest average score with M = 40.22 (SD = 9.27), suggesting that seniors generally view Trinity as academically challenging. CE also shows a high mean with M = 40.11 (SD = 8.70), indicating that students perceive the Trinity environment positively. In contrast, EF, similar to Table 4, has the lowest mean with M = 37.07 (SD = 10.56). The correlation matrix shows positive relationships among all the themes; the strongest association is between AC and EF (r =

0.60). This indicates that students with more academically challenging coursework tend to have more positive experiences with faculty.

Table 5. Descriptive Statistics and Correlations of Academic Themes for Seniors

Construct	Mean	SD	1	2	3	4
1. AC	40.22	9.27	**0.72**			
2. LP	39.82	10.54	0.52	**0.82**		
3. EF	37.07	10.56	0.60	0.52	**0.74**	
4. CE	40.11	8.70	0.25	0.28	0.23	**0.86**

Note: AC = Academic Challenge; LP = Learning with Peers; EF = Experiences with Faculty; CE = Campus Environment.

RQ2. Do changes in Academic Challenge and Experiences with Faculty from freshman to senior year differ between students who participated in study abroad and/or internships and those who did not?

Table 6 presents descriptive statistics and correlations for the engagement indicators AC and EF for both freshmen and seniors. On average, seniors report slightly higher levels of engagement than freshmen in both areas. The mean AC score increases from M = 37.16 (SD = 9.11) for freshmen to M = 40.01 (SD = 8.25) for seniors, while EF increases from M = 31.91 (SD = 9.18) for freshmen to M = 37.19 (SD = 9.47) for seniors. The correlation results suggest positive relationships among the engagement indicators. The strongest relationship appears between AC for freshmen and EF for freshmen (r = 0.52), indicating that freshmen who experience higher academic challenges also tend to report more interactions with faculty. Additionally, EF for seniors is moderately correlated with AC for seniors (r = 0.57). Overall, the correlations are positive and moderate in strength, suggesting that different forms of student engagement tend to occur together.

Table 6. Descriptive Statistics and Correlations of Engagement Indicators

Construct	Mean	SD	1	2	3	4
1. AC F.	37.16	9.11	**0.76**			
2. AC S.	40.01	8.25	0.39	**0.72**		
3. EF F.	31.91	9.18	0.52	0.29	**0.76**	
4. EF S.	37.19	9.47	0.32	0.57	0.36	**0.74**

Note: AC F. = Academic Challenge for freshman, AC S. = Academic Challenge for seniors, EF F. = Experience with Faculty for freshman, EF S. = Experience with Faculty for seniors

Table 7 shows participation in experiential learning activities by class year. The results indicate a substantial difference between freshmen and seniors. Among freshmen, only 13 students (10.92%) reported participating in experiential learning activities, while the vast majority—106 students (89.08%)—reported no participation. In contrast, seniors show much higher engagement, with 85 students (71.43%) reporting participation and only 34 students (28.57%) reporting no participation. This large shift suggests that experiential learning opportunities are much more common later in students' academic careers. As students progress toward their senior year, they are significantly more likely to participate in activities, such as internships, research, or other experiential learning opportunities, which may reflect increased access to these opportunities or program requirements that occur later in college.

Table 7. Experiential Learning Participation by Class Year

Year	Experiential Learning	Count	Percent
Freshman	Yes	13	10.92%
	No	106	89.08%
Senior	Yes	85	71.43%
	No	34	28.57%

Note: Yes = Student did participate in experiential learning activity. No = student did not participate in experiential learning activity.

RQ3. Do perceptions of Campus Environment and Experiences with Faculty differ among the four academic schools (A&H, SS&CE, SSS, NSB)?

Table 8 presents the distribution of respondents across Trinity University's four academic schools for the 2022 and 2025 NSSE survey cohorts. In both years, the largest proportion of respondents came from SSS, followed by SS&CE, NSB, and A&H. The distribution of respondents across schools is relatively consistent between the two survey years, suggesting that the sample composition is comparable across cohorts.

Table 8. Distribution of Respondents by Academic School and Survey Year

Year	School	Count	Percent
2022	A&H	44	9.46%
	NSB	102	21.94%
	SS&CE	130	27.96%
	SSS	189	40.65%
2025	A&H	39	8.71%
	NSB	100	22.32%
	SS&CE	110	24.55%
	SSS	199	44.42%

Note: A&H = Arts & Humanities; SS&CE = Social Sciences & Civic Engagement, SSS = Semmes School of Science, NSB = Neidorff School of Business

Table 9 summarizes the descriptive statistics for the engagement themes of CE and EF by survey year. Mean scores for both engagement themes were higher in 2025 compared to 2022. Specifically, the average CE score increased from M = 39.27 (SD = 9.32) in 2022 to M = 41.52 (SD = 9.56) in 2025, while the average EF score increased from M = 32.42 (SD = 10.59) to M = 35.28 (SD = 11.31). The standard deviations indicate moderate variability in responses across both years.

Table 9. Engagement Theme Means and Standard Deviations by Survey Year

		CE			EF	
Year	**Mean**	**SD**	**Count**	**Mean**	**SD**	**Count**
2022	39.27	9.32	475	32.42	10.59	518
2025	41.52	9.56	444	35.28	11.31	478
Note: CE = Campus Environment; EF = Experiences with Faculty						

Analytical Methods

RQ1. Do student perceptions of Academic Challenge, Learning with Peers, Experiences with Faculty, and Campus Environment change from freshman year to senior year?

The primary objective for this analysis was to better understand how student engagement indicators fluctuate from their freshman to senior year. To address this question, a repeated measures design was employed, leveraging the fact that student observations could be matched across the four years using the **FakeID** column. This approach ensured that the differences observed reflected true student-by-student changes in engagement rather than cohort variations.

The analysis focused on the four overarching themes identified in the Data Cleaning & Reshaping section: AC, LP, EF, and CE. To understand if there was a significant multivariate change in student engagement as a whole across time, a MANOVA was implemented. To evaluate whether the engagement levels changed over time for specific themes, a series of repeated-measure ANOVA tests were employed for each of the four themes. The within-subject factor was Time, with two levels: freshman and senior year. This assessed whether the mean engagement score differed significantly across time.

Once the repeated-measures ANOVA was performed, a secondary analysis was performed at the engagement indicator level to better understand which specific indicators were driving the change within the four themes.

RQ2. Do changes in Academic Challenge and Experiences with Faculty from freshman to senior year differ between students who participated in study abroad and/or internships and those who did not?

To address RQ2, a linear mixed effects model with random slopes was estimated separately for AC and EF. This approach was selected because the repeated-measures structure of the data, where each student contributed to one freshman and one senior observation, violated the independence assumption of standard ANOVA. Linear mixed effects models account for this dependency by modeling individual student variability explicitly through random effects, making them better suited for this analysis than the repeated-measures ANOVA used in RQ1.

Each model included two fixed effects and their interaction. Time (freshman vs. senior) was included as a within-subject factor to capture overall change in engagement across the undergraduate experience. Experiential learning participation (participant vs. non-participant) was included as a between-subject factor to capture baseline differences between the two groups. Critically, the interaction term between time and experiential learning was included to test whether the rate of change in engagement from freshman to senior year differed depending on whether a student participated in internships and/or study abroad. This interaction effect was the primary term of interest for RQ2.

Random slopes for time were also included in each model to allow the effect of time to vary across individual students, reflecting the reality that students may change at different rates over the course of their undergraduate experience. Model estimation was carried out using Restricted Maximum Likelihood (REML). All analyses were conducted in Python.

RQ3. Do perceptions of Campus Environment and Experiences with Faculty differ among the four academic schools (A&H, SS&CE, SSS, NSB)?

A one-way analysis of variance (ANOVA) was conducted for each engagement theme of CE and EF to examine whether student perceptions of engagement differed across academic schools. The analysis used a pooled cross-sectional dataset combining observations from the 2022 and 2025 NSSE survey cohorts. This was important to highlight since descriptive statistics were calculated separately by survey year (Tables 8 and 9) to provide context for the sample composition and engagement levels across cohorts. However, the ANOVA models were estimated on the pooled dataset to evaluate overall differences in engagement across academic schools.

Analytical Results

RQ1. Do student perceptions of Academic Challenge, Learning with Peers, Experiences with Faculty, and Campus Environment change from freshman year to senior year?

A repeated-measures MANOVA was implemented to determine whether the four engagement themes changed significantly from freshman year to senior year as a whole. Using a complete-case sample of 92 students, the multivariate test was statistically significant, Pillai's

Trace = 0.2346, $F(4, 88) = 6.74$, $p < .001$. This result demonstrated that students' engagement experiences changed significantly over time, supporting the notion of an overall longitudinal effect across the four dimensions of engagement.

Mean comparisons between freshman and senior year in Table 10 further clarified the direction of change, indicating that AC and EF increased over time, while LP remained relatively stable and CE declined:

Table 10. Longitudinal Changes in Engagement Themes Freshman to Senior Year

Theme	Freshman Mean	Senior Mean	Mean Change
AC	38.05	40.05	+2.01
LP	39.21	39.67	+0.46
EF	32.80	36.86	+4.06
CE	42.03	39.62	-2.41
Note: AC = Academic Challenge; LP = Learning with Peers; EF = Experiences with Faculty; CE = Campus Environment			

To follow up the MANOVA investigation, repeated measures ANOVAs were conducted to identify which dimensions of the engagement themes contributed to the overall multivariate effect. There was a significant increase in AC from freshman to senior year, $F(1, 103) = 6.13$, $p = .015$. No significant difference was found for LP, $F(1, 108) = 0.73$, $p = .396$. A significant increase was observed in EF, $F(1, 100) = 17.56$, $p < .001$, representing the strongest effect among the four themes. A significant decrease was found for CE, $F(1, 94) = 4.40$, $p = .039$, indicating that students' perceptions of the campus environment, including the quality of their interactions with campus faculty and support staff available to them, were meaningfully lower by senior year than they had been as freshmen.

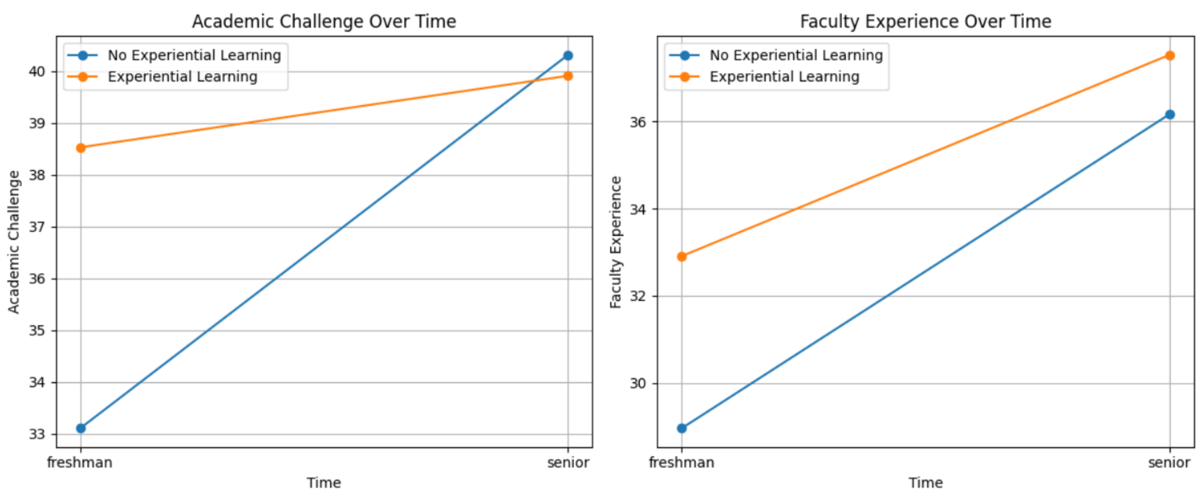
RQ2. Do changes in Academic Challenge and Experiences with Faculty from freshman to senior year differ between students who participated in study abroad and/or internships and those who did not?

The linear mixed effects model for AC converged successfully and fit the data well. The intercept was statistically significant ($\beta = 33.10$, $p < 0.001$), representing the estimated average AC score for non-participants during freshman year. The main effect of time was significant ($\beta = 7.21$, $p < 0.001$), indicating that non-participants experienced a meaningful increase in AC from freshman to senior year. The main effect of experiential learning was also significant ($\beta = 5.42$, $p = 0.004$), suggesting that participants began freshman year with notably higher AC scores than non-participants. Most importantly, the interaction between time and experiential learning was statistically significant ($\beta = -5.83$, $p = 0.003$). This negative interaction indicates that while both groups increased in AC over time, non-participants experienced a steeper rate of growth.

The linear mixed effects model for EF also converged successfully. The intercept was statistically significant ($\beta = 28.96$, $p < 0.001$), representing the estimated average EF score for non-participants during freshman year. The main effect of time was significant ($\beta = 7.21$, $p < 0.001$), indicating meaningful growth in EF for non-participants over time. The main effect of experiential learning was marginally significant ($\beta = 3.95$, $p = 0.039$), suggesting that participants entered freshman year with somewhat higher EF scores than non-participants. However, unlike the AC model, the interaction between time and experiential learning was not statistically significant ($\beta = -2.59$, $p = 0.243$). This indicates that the rate of change in EF from freshman to senior year did not differ meaningfully between participants and non-participants.

Figure 1 displays the estimated marginal means for AC and EF from freshman to senior year, separated by experiential learning participation. In the AC plot, experiential learning participants begin freshman year with notably higher scores but experience a shallower rate of growth, such that non-participants nearly close the gap by senior year — a pattern consistent with the significant negative interaction effect. In the EF plot, participants similarly enter with higher scores and both groups grow at comparable rates through senior year, reflecting the non-significant interaction and parallel trajectories observed in the model.

Figure 1. AC and EF Trajectories by Experiential Learning Participation



RQ3. Do perceptions of Campus Environment and Experiences with Faculty differ among the four academic schools (A&H, SS&CE, SSS, NSB)?

A one-way ANOVA showed no statistically significant differences in engagement across academic schools for either CE or EF. CE returned $F = 0.22$, $p = 0.881$, and EF returned $F = 0.79$, $p = 0.501$. The high p-values for both suggested that students reported similar levels of engagement regardless of academic school. Given this, no further exploration was conducted into the engagement indicators underlying each theme.

Analytical Discussion

RQ1. Do student perceptions of Academic Challenge, Learning with Peers, Experiences with Faculty, and Campus Environment change from freshman year to senior year?

The MANOVA result shows that students' engagement experiences change substantially over time. The most clear trend is the substantial increase in EF, which showed the largest mean change and the strongest statistical significance. As students progress through their undergraduate career, their interactions with faculty become more frequent and consequential to their academic and professional development. Given the structure of higher education, faculty engagement becomes more frequent in upper-division coursework. Furthermore, Trinity's smaller class sizes, major-specific courses, and greater academic maturity likely contribute to these enhanced interactions. This outcome is very positive, as previous research consistently links faculty engagement with improved learning outcomes and higher student satisfaction.

Beyond faculty engagement, the increase in AC indicates that students perceive their academic experience as becoming more rigorous over time. As students advance, they encounter coursework that demands higher-level skills, deeper analytical thinking, and more complex problem-solving in ambiguous environments. This suggests that students are developing the critical competencies targeted by higher education, and the increased challenge reflects positive growth in academic capability.

In contrast, LP remained stable over time. This suggests that collaborative learning behaviors and peer interactions are established early and persist throughout college. It may also indicate that institutions provide consistent opportunities for peer engagement at all academic levels.

However, CE declined significantly from freshman to senior year. This suggests that students may become more critical of their institutional environment over time, especially regarding the quality of interactions. Students may enter college with optimistic expectations that are not fully sustained, or they may become more aware of limitations in support, communication, or administrative processes as they gain experience.

RQ2. Do changes in Academic Challenge and Experiences with Faculty from freshman to senior year differ between students who participated in study abroad and/or internships and those who did not?

The results for RQ2 present a nuanced picture of how experiential learning participation relates to engagement trajectories over the undergraduate experience. For AC, the significant time-by-experiential learning interaction reveals a convergence pattern: students who did not participate in internships or study abroad started with considerably lower AC scores as freshmen but grew at a steeper rate, largely catching up to participants by senior year. This finding suggests that experiential learning participation may serve as an early differentiator in academic engagement, but that non-participants are not permanently disadvantaged. One interpretation is that non-participants experience greater academic growth through other means over the course of their undergraduate career, or that the academic demands of later coursework produce similar levels of challenge regardless of experiential participation.

For EF, the absence of a significant interaction suggests that experiential learning participation does not meaningfully alter how students' faculty engagement evolves over time. Both groups showed substantial growth in EF from freshman to senior year, which aligns with the broader finding from RQ1 that EF increases significantly across the undergraduate experience. The non-significant interaction implies that this growth is largely independent of whether a student participated in high-impact practices.

Taken together, these findings carry practical implications for TUIR and institutional leadership. The convergence in AC scores suggests that students who miss out on early experiential learning opportunities are not necessarily at a long-term disadvantage in terms of academic engagement, though the source of their later growth warrants further investigation. At the same time, the consistent growth in EF across both groups reinforces the importance of faculty engagement as a broadly shared developmental outcome, one that does not appear to be contingent on participation in specific high-impact practices. These results suggest that TUIR's continued attention to EF at the institutional level is well-placed, while the differential early trajectory in AC may warrant targeted outreach to encourage earlier participation in experiential learning opportunities.

RQ3. Do perceptions of Campus Environment and Experiences with Faculty differ among the four academic schools (A&H, SS&CE, SSS, NSB)?

While school-level segmentation is a common way to break down data analytically, it is not a particularly useful lens for CE and EF in this scenario as students report similar levels of engagement regardless of their academic school. As a result, this helps narrow the set of variables that are worth focusing on moving forward.

For TUIR, this gives clear direction. For instance, instead of continuing to examine school-level comparisons for these outcomes, future analysis should focus on areas that have meaningful variation, specifically changes over time and participation in high-impact practices, as seen in RQ1 and RQ2. Furthermore, the consistency across schools suggests that any efforts to improve CE or EF can be approached at the institutional level, without the need to target specific academic divisions.

Conclusion

Taken together, the findings from this study point to a consistent institutional story: student engagement at Trinity develops meaningfully over time, and that development is most pronounced in AC and EF. The lack of significant school-level differences in CE and EF further reinforces that these trends are institution-wide phenomena rather than the result of any academic division.

These results help clarify where future research energy is best directed. Rather than continuing to examine school-level segmentation for CE and EF, subsequent analyses should focus on understanding the mechanisms behind the convergence observed in AC. More specifically, future research should investigate what academic pathways drive growth among students who do not participate in early experiential learning. Additionally, the decline in CE warrants deeper investigation, particularly into whether specific student subgroups or touchpoints drive that dissatisfaction. Longitudinal tracking of these patterns across future NSSE administrations would allow TUIR to assess whether institutional interventions are producing measurable improvements over time.